

Data Structures & Algorithms

BCA — Semester III — 2022 — Answer Sheet
Subject Code: CACS201

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. What is abstract data type? Convert $a \cdot b \cdot c - d + e / f / (g + h)$ into postfix expression using stack.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is linked list? Describe types of linked list. Write an algorithm to insert and delete node from beginning of doubly linked list.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Describe Prim's algorithm to solve MST problem with suitable illustration.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is the limitation of linear queue over circular queue? Write an algorithm to insert and delete node in circular queue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is hashing? Describe the types of collision resolution techniques with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define divide and conquer algorithm. What is binary search? Write an algorithm to search an item using binary search with suitable illustration.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**8. What is minimax algorithm? Create Huffman Tree and calculate Huffman code for the following characters along with their frequencies using Huffman algorithm. Characters : A, E, I, O, U, S, T
Frequencies: 10, 15, 12, 3, 4, 13, 1**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is stack? List the application of the stack. Write an algorithm to perform PUSH and POP operation in stack. Describe linked list implementation of stack operations.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is external sorting? Explain heap sort algorithm and trace it to sort the data: 82, 90, 10, 12, 15, 77, 55, 23, 25, 32

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Differentiate between BST and AVL tree. Given the following AVL Tree: Draw the resulting BST after 5 is removed, but before any rebalancing takes place. Label each node in the resulting tree with its balance factor. Replace a node with both children using an appropriate value from the node's left child.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.